

Daily Progress & Next Steps Plan

Constraint-Interaction Metric for Symbolic Regression

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July 10, 2026

Part 1: Summary of Today's Work (Constraints C1–C4)

Today's focus was implementing the four core constraint predicates in `src/constraints.py`, completing the `expr_generator` $\rightarrow$ `metric` $\rightarrow$ `constraints` pipeline.

- **C1 – Structural Constraint:** Implemented `make_c1_structural` enforcing two sub-rules: (a) maximum nested trig depth of 1 (`sin(cos(x))` rejected; parallel siblings accepted); (b) maximum flattened `Add/Mul` arity of 3 (2 consecutive binary ops at limit). Handled SymPy's automatic flattening of repeated additions by checking `len(node.args)` rather than tree depth.
- **C2 – Depth Constraint:** Implemented `make_c2_depth` using a recursive `.args`-walk depth. Formally documented the divergence between SymPy depth and generator string depth (e.g., `-x` $\rightarrow$ `Mul(-1, x)`, depth 1). The bias is bounded ($\leq +2$ per affected node) and measured empirically by `test_depth_divergence_measured` (mean gap: 0.42, max: 3). Rejected the opt-in passthrough alternative to preserve the `Expr` $\rightarrow$ `bool` purity contract.
- **C3 – Operator Whitelist:** Implemented `make_c3_operator` via `preorder_traversal`, skipping leaf types (`Symbol`, `Number`, etc.) and rejecting any non-leaf node whose type is outside the frozenset of permitted classes. The single `_OP_CLASS_MAP` dict (Bug #4 fix) correctly expands `"-"` to both `Add` and `Mul`. Empirically confirmed $\rho(C_3) \approx 0.23$, which is in the expected 0.3–0.6 band (a value near 1.0 would indicate a bug).
- **C4 – Positivity (Non-Negativity):** Implemented `make_c4_positivity` with a vectorised `lambdify` fast path and a pointwise `.subs()` fallback. Predicate is $f(x, y) \geq 0$ (not > 0) per Research Plan §5; boundary case $f = 0$ returns `True`. RNG seed is read exclusively from `config['constraints']['positivity']['rng_seed']` (Bug #2 fix).
- **Testing:** 29 new tests added across `TestC1Structural`, `TestC2Depth`, `TestC3OperatorWhitelist`, `TestC4Positivity`, and `TestBuildConstraints`. Includes mandatory boundary-locking tests and the empirical divergence measurement for Handbook §12. Full suite: **60 tests, all passing**.
- **Integration Smoke Test:** `DensityEstimator.estimate(build_constraints(cfg), N=1000)` returned $\rho = [0.404, 0.987, 0.227, 0.231]$ – all within the expected ranges from the plan.
- **Documentation:** Updated `DESIGN_CONTEXT.md` (§6.2 marked Implemented; four new Known Gotchas), `API.md` (`src.constraints` module section), `config.yaml` (`rng_seed`: 0 + clarifying comment), and `LOGBOOK.md` (Entry 06).

Part 2: Strict Plan for Tomorrow

With constraints C1–C4 complete, tomorrow’s work will close the pipeline by implementing `src/experiments.py` and running the full metric computation.

Phase 1: Experiment Runner (`src/experiments.py`)

Build the experiment orchestration layer following `DESIGN_CONTEXT.md` §9.

- **Reproducibility Logging:** Snapshot Git hash, config SHA-256, Python version, library versions, and timestamp to `results/<run_id>/metadata.json` before each run starts.
- **Multi-seed Runs:** Iterate over `config.experiments.seeds` and `config.monte_carlo.N`, calling `DensityEstimator.estimate(build_constraints(cfg), N)` for each seed.
- **Matrix + CI:** Compute `InteractionMatrix` and `BootstrapCI` ($B = 1000$) per run; store the $M(i, j)$ array and 95% CI bounds to JSON.
- **Append-Only Results:** Each run writes to a new `run_id` directory; no overwriting of prior results (`DESIGN_CONTEXT` §6.4).

Phase 2: Full Pipeline Run

Execute the complete pipeline with $N = 100,000$ and the five seeds from `config`, producing the definitive ρ_i and $M(i, j)$ tables for the Research Plan §7 results section.

Phase 3 (If Time Permits): C5 – Dimensional Consistency

Implement `make_c5_dimensional` using a manual dimension-algebra tree walk with `fractions.Fraction` exponent tracking (no `sympy.physics.units`) and enable it via `build_constraints(cfg, include_c5=True)`.